$$0 = 1 - 1 = -1 + 1 = 0$$

From Elementary Arithmetic to Intrinsic Skein-Lasagna

Keyao Peng

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Question

Can you prove $0 = 0$ in a nontrivial way?

Test your math level

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Hopf fibration

Hopf fibration is a map defined by

$$H : \mathbb{S}^3 \subset \mathbb{C}^2 - \{0\} \to \mathbb{CP}^1 \cong \mathbb{S}^2, (z_1, z_2) \mapsto [z_1, z_2]$$

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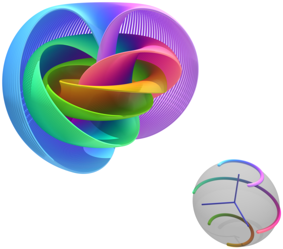
For any point $x \in \mathbb{S}^2$, preimage $H^{-1}(x)$ is a circle $\mathbb{S}^1$. We can visualize this map by identifying $\mathbb{R}^3 \cong \mathbb{S}^3 - \{\infty\}$, so that the Hopf fibration appears as a map from $\mathbb{R}^3$ to $\mathbb{S}^2$:



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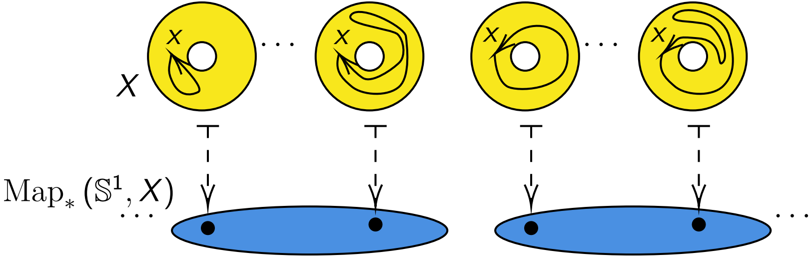

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Let us now study the case $X = \mathbb{S}^n$.

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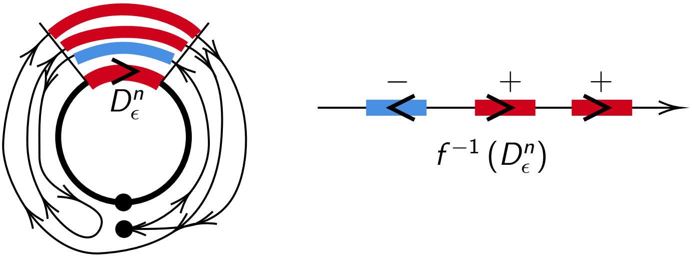
D^n_ϵ
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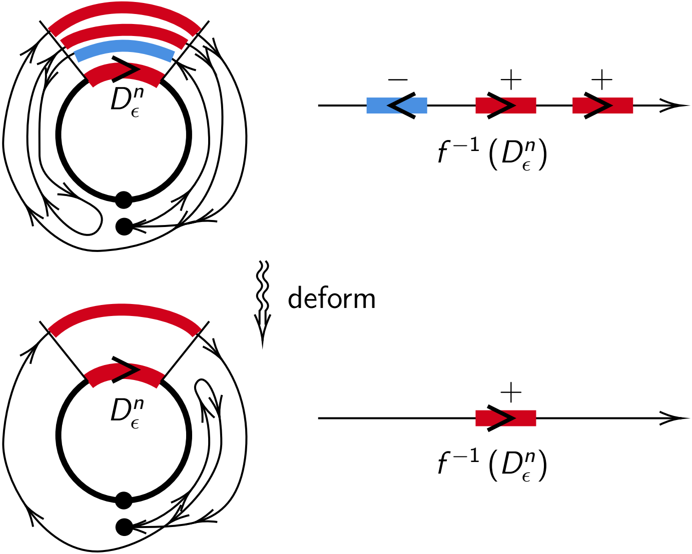
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Homotopy groups of spheres

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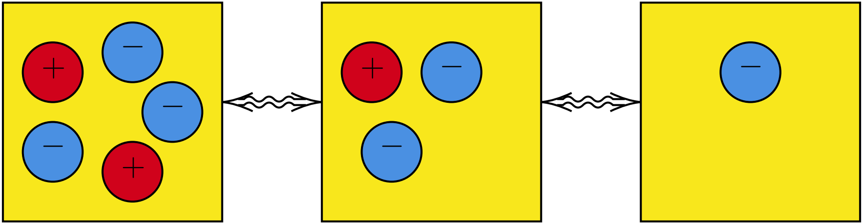

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This gives an isomorphism to the configuration space of disks.

$$\mathrm{End}_*(\mathbb{S}^n) \cong \mathrm{Conf}^{fr}(\mathbb{R}^n) = \bigcup_{j,k} \mathrm{Emb}(\bigsqcup_j D^n_+ \sqcup \bigsqcup_k D^n_-, \mathbb{R}^n)$$

Therefore, $\pi_0 \mathrm{End}_*(\mathbb{S}^n) \cong \mathbb{Z}, (j,k) \mapsto j - k$

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Higher homotopy groups of spheres

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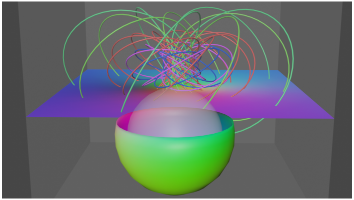
$$\mathrm{Map}_*(\mathbb{S}^{n+1}, \mathbb{S}^n) \cong \mathrm{Map}_*(\mathbb{S}^1, \mathrm{End}_*(\mathbb{S}^n)) \to \mathrm{Map}_*([0,1], \mathrm{End}_*(\mathbb{S}^n))$$

Then H corresponds to a path (that is, a homotopy) from $0 \in \mathrm{End}_*(\mathbb{S}^2)$ back to itself.

I made an animation to visualize this deformation.



Visualize H as deformation

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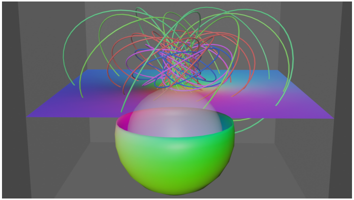

We can also visualize it as a movement in the configuration space. We will show it later.

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We use the **Grothendieck group** $K_0(\mathrm{FinSet})$:

$$K_0(\mathrm{FinSet}) = \{(X, Y) \in (\mathrm{FinSet}_{/\cong})^2\}/\{(X \sqcup Z, Y \sqcup Z) \sim (X, Y)\}$$

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But what is $K_i(\mathrm{FinSet})$ for $i > 0$?

Space of finite sets

Instead of thinking the set of "finite sets", we now consider the space of them $F = \mathrm{FinSet}^{\cong}$: every point $x \in F$ corresponds to a finite set, and a path between to points corresponds to an isomorphism $f : x \xrightarrow{\cong} y$ of two finite sets.

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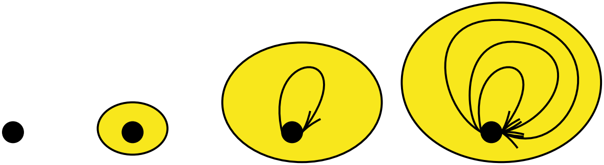

$\emptyset$ $\quad |x| = 1 \quad |x| = 2 \quad |x| = 3 \quad \ldots$

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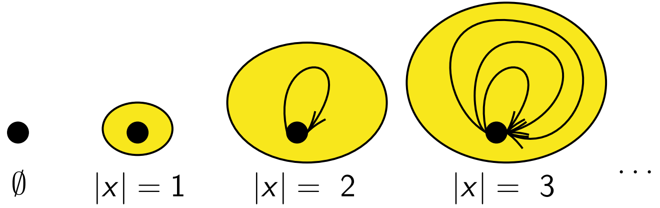

- $\pi_0(F) = \mathrm{FinSet}_{/\cong} \cong \mathbb{N}$.
- $\pi_1(F, x) = \mathrm{S}_{|x|}$, the permutation group.
- $\pi_i(F, x) = \{0\}$ for $i > 1$ (one can never deform an isomorphism of set to another).

K-space (spectrum) of finite sets

We now want to add "negative" elements to F. To do this, we consider the space $K(F)$ of ordered finite sets whose elements are marked by $+$ or $-$. We identify signed sets with the same value $\|X\| = |X_+| - |X_-|$, and these identifications are represented by paths.

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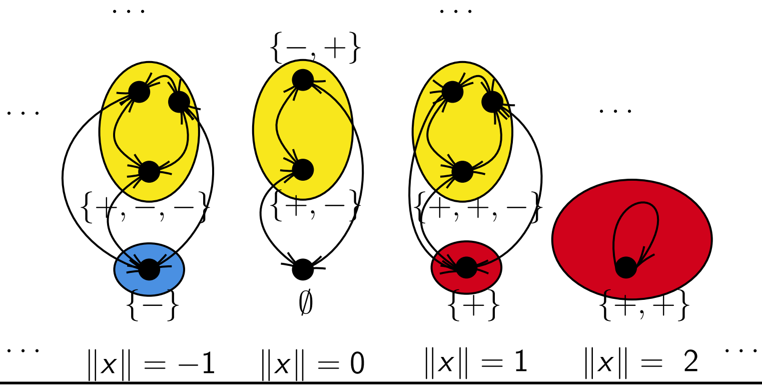

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Answer

We can prove $0 = 0$ nontrivially with $h : 0 = 1 - 1 = -1 + 1 = 0$!

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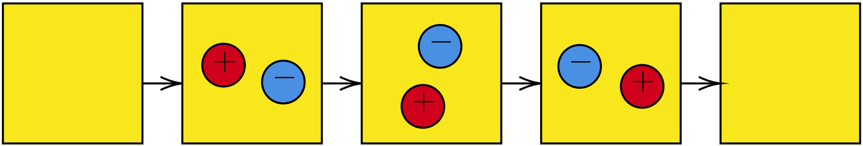

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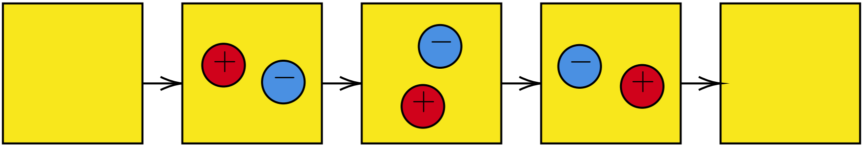
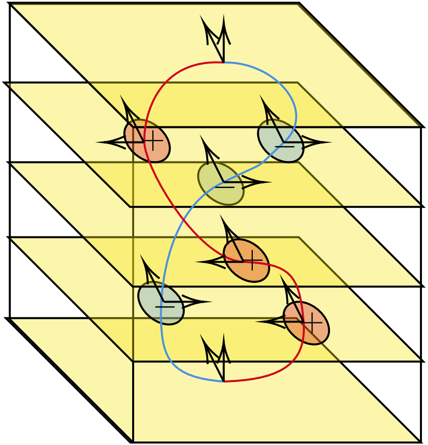


New Question

Can we find a geometric interpretation of the general K-groups of a field $K_i(k)$?

What about cobordism?

If we place this motion in 3-dimensional space, we obtain a circle:



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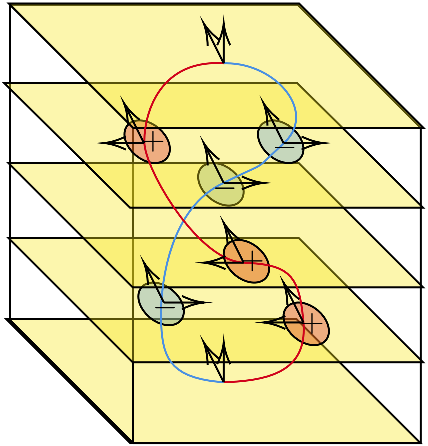

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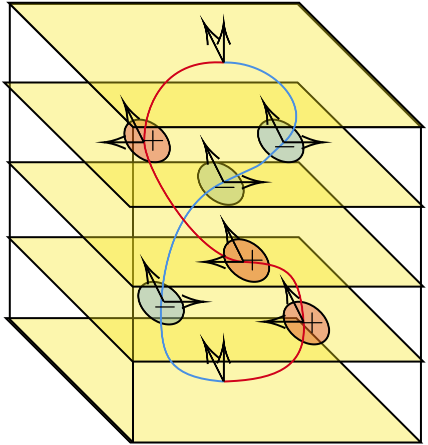

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$$\Omega_i^{fr} \cong \pi_i^s(\mathbb{S})$$

Therefore, this framed circle is nontrivial and generates Ω_1^{fr}.

Intrinsic Skein-Lasagna

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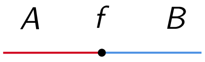
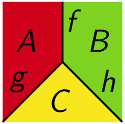
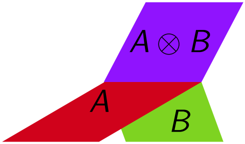
- 0-codimensional strata are labeled by objects; for $i > 0$, i-codimensional strata are labeled by i-morphisms,
- At the branches of the treefold, we mark how pieces are combined using the monoidal product $\otimes$.



$$A \xrightarrow{f} B$$



$$\begin{tikzcd} A \arrow[rr, "f"] \arrow[dr, "g"'] & & B \\ & C \arrow[ur, "h"'] \arrow[u, phantom, "\Rrightarrow"] & \end{tikzcd}$$



Relation with "Skein-lasagna module"

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Cobordism of lasagna

As expected, we define a cobordism relation between L_1 and L_2 if there exists a lasagna W with boundary such that $\partial W = L_1 \sqcup -L_2$. Finally, we define the nth $(\mathcal{C}, \otimes)$-**cobordism group** $\Omega_n^{\mathcal{C},\otimes}$ to be the abelian group generated by n-lasagnas modulo these cobordism relations.

As an example, consider $\Omega_n^{V_k,\oplus}$, where $V_k = \mathrm{FinVect}_k^{\cong}$ is the category of finite-dimensional vector spaces with all isomorphisms as morphisms. In this case, one can think of a lasagna as a treefold/manifold equipped with a k-local system.

$(V_k, \oplus)$-Cobordism Group

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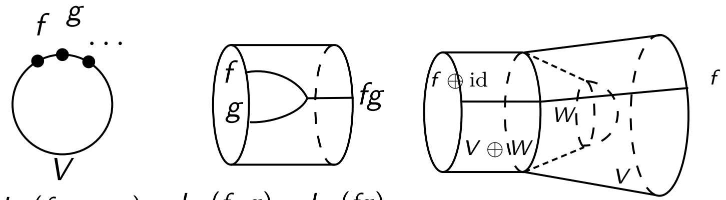
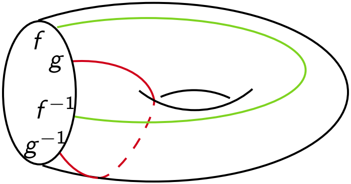
- Clearly, $\Omega_0^{V_k,\oplus} \cong \mathbb{Z}$, classified by dimension of vector space.
- $\Omega_1^{V_k,\oplus}$ is generated by circles $L_V(f)$ labeled by a vector space V and a matrix $f \in \mathrm{GL}_V(k)$. By stabilizing with identity matrices, we obtain a homomorphism $L : \mathrm{GL}_\infty(k) \twoheadrightarrow \Omega_1^{V_k,\oplus}$.



$L_V(f, g, \dots)$ $\quad$ $L_V(f, g) \sim L_V(fg)$ $\quad$ $L_{V \oplus W}(f \oplus \mathrm{id}) \sim L_V(f)$

We have one more relation coming from surfaces:



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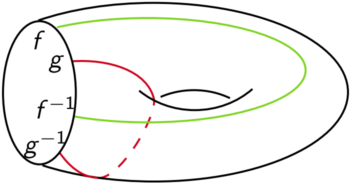

$$L([f,g]) \sim 0$$

Notice that $k^\times \cong \mathrm{GL}_\infty(k)/[\mathrm{GL}_\infty(k), \mathrm{GL}_\infty(k)]$. In fact, one can show that the induced homomorphism $L : k^\times \xrightarrow{\cong} \Omega_1^{V_k,\oplus}$ is an isomorphism. The lasagna induced by the Hopf map H is then given by $L(-1)$.

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Generalized Pontryagin–Thom Isomorphism (Claim)

For all $n \geq 0$, we have the isomorphism

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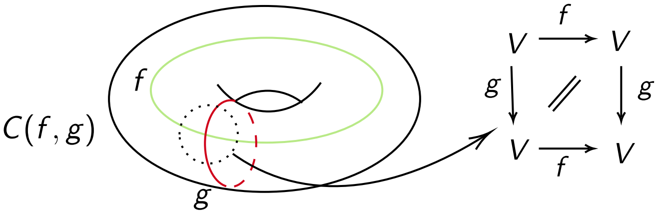
This answers the new question: it suggests that we can view lasagnas as elements of K-groups, providing a bridge from difficult problems in algebra to difficult problems in geometric topology.
So what does this tell us when $n = 2$?

$\Omega_2^{V_k,\oplus}$ and $K_2(k)$

For matrices $f, g \in \mathrm{GL}(k)$ such that $[f, g] = 1$, one can define the following 2-lasagna $C(f, g)$:



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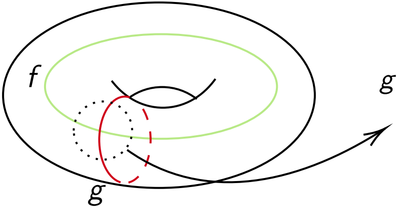

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "g"'] \arrow[dr, phantom, "/\!/"] & V \arrow[d, "g"] \\ V \arrow[r, "f"'] & V \end{tikzcd}$$

Recall that $K_2(k) = k^\times \otimes k^\times / \langle a \otimes (1-a) \rangle$, which means that we should find a 3-lasagna, called Ghys' lasagna, that bounds $C(a, 1-a)$. In other words, we can encode a proof or computation by a lasagna.

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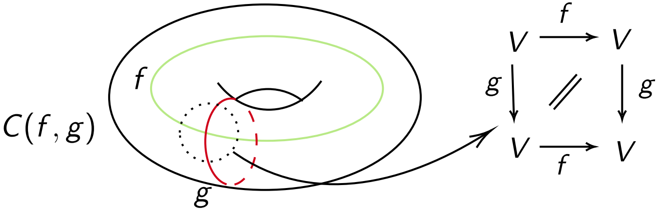

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I do not yet have Ghys' lasagna in hand (it has Heegaard genus less than 26), but I do have an easier example.

For $i \neq j$, let the elementary matrices $e_{ij}(a)$ be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

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Notice that $e_{ij}(a)^{-1} = e_{ij}(-a)$, and these matrices satisfy the Steinberg relations:

$$[e_{ij}(a), e_{kl}(b)] = \begin{cases} e_{il}(ab), & \text{if } j = k, \\ \mathbf{1}, & \text{otherwise.} \end{cases}$$

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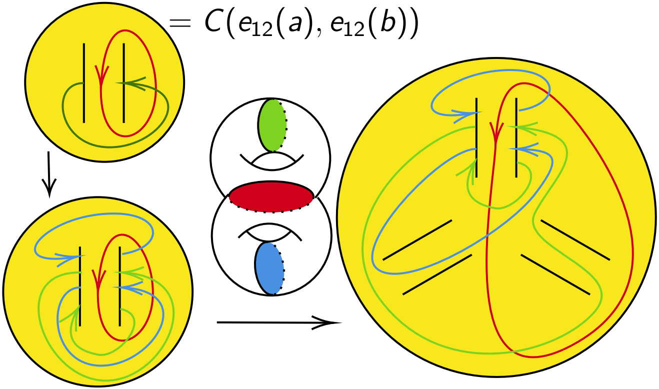
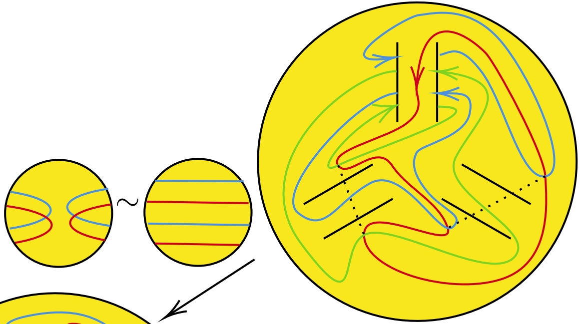
Using these relations and the Hall–Witt identity, we obtain the following nontrivial proof that $[e_{12}(a), e_{12}(b)] = \mathbf{1}$:

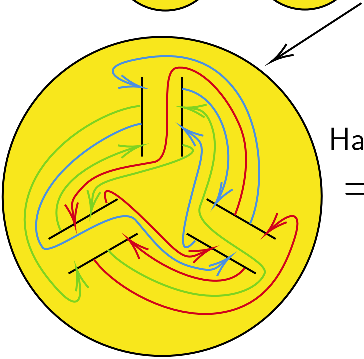
$$\begin{gathered} [e_{12}(a), e_{12}(b)] = [[e_{13}(a), e_{32}(1)], e_{12}(b)] \\ = [e_{32}(1), [e_{13}(-a), e_{12}(-b)]]^{e_{13}(a)}[e_{13}(a), [e_{32}(-1), e_{12}(-b)]]^{e_{32}(-1)} \\ = [e_{32}(1), \mathbf{1}]^{e_{13}(a)}[e_{13}(a), \mathbf{1}]^{e_{32}(-1)} = \mathbf{1} \end{gathered}$$

Following this blueprint, we obtain the following Heegaard splitting to construct a lasagna:

$$= C(e_{12}(a), e_{12}(b))$$

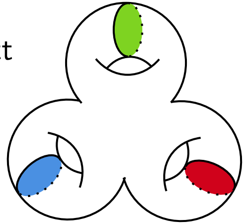






Hall-Witt

$= \partial$



Thank you! I hope you enjoyed and gained some new insights.